```
#string
str1 = 'Syed'
str2= "Uzair"
str3 = '''Hussain'''
print(str1)
print(str2)
print(str3)
```

```
Syed
Uzair
Hussain
```

```
str_text = "this is uzair's"
print(str_text)
str_text
```

```
this is uzair's
'this is uzair's'
```

```
#lenght
len(str1)
```

```
4
```

```
str_4 = "this is syed uzair hussain"
len(str_4)
```

```
26
```

```
name_1 ="Lacking inspiration?Here are the latest recommendations received on Superprof."
print(name_1)
len(name_1)
```

```
Lacking inspiration?Here are the latest recommendations received on Superprof.
78
```

```
name = input("enter your name:")
len(name)
```

```
enter your name:uzair
5
```

```
name_1 ="Lacking inspiration?Here are the latest recommendations received on Superprof."
len(name_1)
name_1
```

```
'Lacking inspiration?Here are the latest recommendations received on Superprof.'
```

```
#indexing/ escape sequence character
class_1 = "This is python class for  PGD students. \nThere are 70 students"
print(class_1)
```

```
This is python class for  PGD students.
There are 70 students
```

```
class_1 = "This is python \t class for  PGD students.There are 70 students"
print(class_1)
```

```
This is python   class for  PGD students.There are 70 students
```

```
a = "NED Academy"
#N=0 E=1 D= 2.....
b = 400
```

```
sum = 2 + 2
r = print(sum)
```

```
4
```

```
text = "Python"
text[0]
```

```
'P'
```

```
text[3]
```

```
'h'
```

```
#SLICING
Job = "Think of the beauty is joy forever"
Job[1:7]#starting index: ending index
```

```
'hink o'
```

```
Job[0:7]
```

```
'Think o'
```

```
Job[4:]
```

```
'k of the beauty is joy forever'
```

```
Job[:10]
```

```
'Think of t'
```

```
Job[:]
```

```
'Think of the beauty is joy forever'
```

```
#Negative indexing
text_1 = "Seven Stages of life"
text_1[-1]
```

```
'e'
```

```
text_1[-7:]
```

```
'of life'
```

```
text_1[3] = "a" #immutable
```

```
---------------------------------------------------------------------------
TypeError                                 Traceback (most recent call last)
/tmp/ipykernel_1308/3136278411.py in <cell line: 0>()
----> 1 text_1[3] = "a"

TypeError: 'str' object does not support item assignment
```

Next steps: Explain error

```
text_1 = "Seven Stages of life"
text_1[6:12] = "Level"
```

```
---------------------------------------------------------------------------
TypeError                                 Traceback (most recent call last)
/tmp/ipykernel_1308/4212643861.py in <cell line: 0>()
      1 text_1 = "Seven Stages of life"
----> 2 text_1[6:12] = "Level"

TypeError: 'str' object does not support item assignment
```

Next steps: Explain error

```
text_1 = "Seven Stages of life"
text_1.endswith("python")
```

```
False
```

```
text_1.endswith("e")
```

```
True
```

```
text_1.endswith("E")
```

```
False
```

```
text_1 = "seven Stages of life"
text_1.capitalize()
```

```
'Seven stages of life'
```

```
t = "seven Stages of life Stages"
t = t.replace("Stages","Levels")
```

```
t
```

```
'seven Levels of life Levels'
```

```
text_1
```

```
'seven Stages of life'
```

```
t1 = "seven Stages of life Stages"
t2 = t1.replace("Stages","Levels")
```

```
t1
```

```
'seven Stages of life Stages'
```

```
t2
```

```
'seven Levels of life Levels'
```

```
t1 = "seven Stages of life Stages"
t2 = t3.replace("Stages","Levels")
```

```
---------------------------------------------------------------------------
NameError                                 Traceback (most recent call last)
/tmp/ipykernel_1308/2437886363.py in <cell line: 0>()
      1 t1 = "seven Stages of life Stages"
----> 2 t2 = t3.replace("Stages","Levels")

NameError: name 't3' is not defined
```

Next steps: Explain error

```
s = "Ali Haider"
s.find("Haider")
```

```
4
```

```
s = "Ali Haider"
s.find("syed")
```

```
-1
```

```
s = "Ali Haider"
s.find("i")
```

```
2
```

```
u = "This is python class with python teacher"
u.count("python")
```

```
2
```

```
u = "This is python class with python teacher"
u.count(" ")
```

```
6
```

```
#Conditional Statements (Identation)
#if condition: (True/False)
#  statement
#else:
#  statement
```

```
#To verify the age limitation for voting
age = 10
if age >= 18:
  print("You are eligible for voting")
else:
  print("You are not eligible for voting")
```

```
You are not eligible for voting
```

```
#To check the Temperature
temperature = float(input("Enter your body temperature:"))
if temperature >= 37.5:
  print("Temperature is high")
else:
  print("Temperature is normal")
```

```
Enter your body temperature:40
Temperature is high
```

```
#
temperature = 30
if temperature > 25:
  print("It is warm")
else:
  print("Drink water")
print("Program finished")
print("it is very supportive")
```

```
It is warm
Program finished
it is very supportive
```

```
temperature = int(input("Enter your water temperature: "))
if temperature > 25:
  print("It is warm")
else:
  print("Drink water")

print("Program finished")
print("it is very supportive")

marks = int(input("Enter your marks: "))
if marks > 40:
  print("You are pass")
else:
  print("Your are failed")
```

```
Enter your water temperature: 25
Drink water
Program finished
it is very supportive
Enter your marks: 35
Your are failed
```

```
#For multiple conditions we use elif(else-if)
#if condition:
# statement

#elif condition:
# statement

#elif condition:
# statement

#else:
# statement
```

```
# make a program to analysis the grade of students
marks = float(input("Enter your marks(it should be between 1 - 100): "))
if marks >= 60:
  print("You are passed")
else:
  print("You are failed")



if marks >= 80:
  print("Grade A*")
elif marks >= 70:
  print("Grade A")
elif marks >= 60:
  print("Grade B")

print(marks)
```

```
Enter your marks(it should be between 1 - 100): 25
You are failed
25.0
```

```
#and or (true and true) = true | (true or false) = true
#check age and citizenship
age = int(input("Enter your age"))
citizen = input("Are you a citizen?(yes/no): ")
if age >= 18 and citizen == "yes":
  print("You are eligible for voting")
else:
  print("You are not eligible for voting")
```

```
Enter your age25
Are you a citizen?(yes/no): no
You are not eligible for voting
```

```
#check  Email and password
email = input("Enter your email: ")
password = input("Enter your password: ")
if email == "syeduzairhussain91@gmail.com" and password == "python123":
  print("Login successful")
else:
  print("Invalid credentials")
```

```
Enter your email: syeduzairhussain91@gmail.com
Enter your password: python123
Login successful
```

```
#check the number range
#admission eligiblity
#weeekend checkery
#Odd/Even number checker
#Employee Bonus Eligibility
#Discount Eligibility
#Emergency Alert
```